

Probability Theory

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Solve the following problem. Solution is due on September 7, 2026.

Box 1: Application Example (Was it the burglar?)

Mary lives in San Francisco. One afternoon, she is driving back home and receives a phone call from her neighbor Jane. She told her that her house alarm was set off (A). While driving, she also heard on the radio (R) that a small earthquake (E) hit the city. Small earthquakes sometimes activate the alarm, and perhaps this is why it was sounding. To solve this problem define the random variables $R(R) = 1, R(R^c) = 0, A(A) = 1, A(A^c) = 0, E(E) = 1, E(E^c) = 0$, and $B(B) = 1, B(B^c) = 0, J(J) = 1$, and $J(J^c) = 0$. For questions 1-3, assume that $p(r, a, b, e) = p(e)p(r|e)p(a|e, b)$. For questions in item 4, assume that $p(r, a, b, e, a) = p(e)p(r|e)p(a|e, b)p(j|a)$.

1. Draw the corresponding graphical models and write the numeric values of the probability mass $p(r, a, b, e)$ into a table.
2. What is the probability that a burglar (B) broke into the house?
3. What is the probability that a burglar broke into the house given that the alarm was set off?
4. What is the probability that a burglar broke into the house given that the alarm was set off and a small earthquake hit the city?
5. Compute the table associated with $p(a, e, r, b)$.
6. Assume that there is a probability that Jane may not have listened to the alarm (J^c).
 - (a) What is the probability that a burglar (B) broke into the house?
 - (b) What is the probability that a burglar broke into the house given Jane heard the alarm beeping?
 - (c) What is the probability that a burglar broke into the house given Jane did not hear the alarm beeping?
 - (d) What is the probability that a burglar broke into the house, given that Jane heard the alarm sound and that a small earthquake hit the city?

For questions 1-3, the factors of the probability distribution corre-

spend to the following tables.

$b \quad p(b)$		
B ^c	0	$b_1 = .9$
B	1	$b_2 = .1$

$e \quad p(e)$		
E ^c	0	$e_1 = 0.95$
E	1	e_2

$r \quad p(r)$		
R ^c	0	$c_1 = 0.05$
R	1	c_2

$b \quad e \quad p(b, e)$				
B ^c	E ^c	0	0	$b_1 e_1$
B ^c	E	0	1	$b_1 e_2$
B	E ^c	1	0	$b_2 e_1$
B.	E	1	1	$b_2 e_2$

$r \quad e \quad p(r e)$				
R ^c	E ^c	0	0	$f_1 = 0.99$
R ^c	E	0	1	$f_2 = 0.01$
R	E ^c	1	0	f_3
R.	E	1	1	f_4

$a \quad b \quad e \quad p(a b, e)$						
A ^c	B ^c	E ^c	0	0	0	q_1
A ^c	B ^c	E	0	0	1	q_2
A ^c	B	E ^c	0	1	0	q_3
A ^c	B	E	0	1	1	q_4
A	B ^c	E ^c	1	0	0	q_5
A	B ^c	E	1	0	1	q_6
A	B	E ^c	1	1	0	q_7
A	B	E	1	1	1	q_8

$$\begin{aligned}
 q_1 + q_5 &= 1, q_2 + q_6 = 1, \\
 q_3 + q_5 &= 1, q_4 + q_8 = 1, \\
 b_1 + b_2 &= 1, e_1 + e_2 = 1, \\
 c_1 + c_2 &= 1, \\
 f_1 + f_3 &= 1, \\
 f_2 + f_4 &= 1.
 \end{aligned}$$

For questions in 4, consider the previous tables as well as the following table:

$j \quad a \quad p(j a)$				
J ^c	A ^c	0	0	$g_1 = 0.999$
J ^c	A	0	1	$g_2 = 0.2$
J	A ^c	1	0	g_3
J.	A	1	1	g_4

The joint distribution $p(r, a, e, b) = p(e)p(r|e)p(a|e, b)$ is

				r	a	b	e	$p(r, a, b, e)$
R^c	A^c	B^c	E^c	0	0	0	0	p_1
R^c	A^c	B^c	E	0	0	0	1	p_2
R^c	A^c	B	E^c	0	0	1	0	p_3
R^c	A^c	B	E	0	0	1	1	p_4
R^c	A	B^c	E^c	0	1	0	0	p_5
R^c	A	B^c	E	0	1	0	1	p_6
R^c	A	B	E^c	0	1	1	0	p_7
R^c	A	B	E	0	1	1	1	p_8
R	A^c	B^c	E^c	1	0	0	0	p_9
R	A^c	B^c	E	1	0	0	1	p_{10}
R	A^c	B	E^c	1	0	1	0	p_{11}
R	A^c	B	E	1	0	1	1	p_{12}
R	A	B^c	E^c	1	1	0	0	p_{13}
R	A	B^c	E	1	1	0	1	p_{14}
R	A	B	E^c	1	1	1	0	p_{15}
R	A	B	E	1	1	1	1	p_{16}